Approximate Computing		
Paderborn University		SoSe 2026
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1 Objective

Get familiar with the application of approximate computing in Machine Learning (ML) algorithms. You will apply the requested approximation techniques on a CNN network to identify their impacts on the performance of the given CNN.

2 Dataset

Throughout this section, we will perform experiments on the MNIST dataset. The ultimate goal is to build a python-based digit recognition system. You may find further information about MNIST in the previous tutorial (Lab Module-1: Tutorial).

3 Target network

The target neural network for the given task is LeNet 5. LeNet 5 is a convolutional neural networks that contains 2 convolution layers followed by 3 fully connected layers. The architecture of the network has been shown in the Fig. 1.

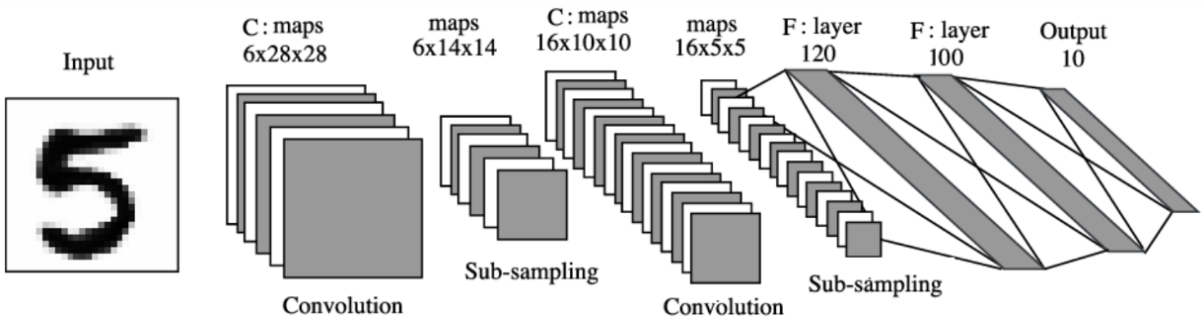


Figure 1: Target CNN

The LeNet 5 contains the following layers

- Convolution: 6 feature maps
- Pooling: 2*2
- Convolution: 16 feature maps
- Pooling: 2*2
- Fully connected layer: 120 neurons
- Fully connected layer: 100 neurons
- Fully connected layer(output layer): 10 neurons

All the layers use the ReLU activation function except the output layer which utilizes softmax to compute the class probabilities.

4 Approximation tasks

We want to apply a bunch of approximation techniques on the given model. This includes pruning, and data compression techniques.

4.1 Weight pruning

Task 1: In the tutorial, weight pruning was applied only to a single fully connected layer. In this task, apply weight pruning on all fully connected layers of the LeNet-5 model using two strategies:

- Random pruning: randomly remove weights in a uniform manner.
- Magnitude-based pruning: remove the weights with the smallest absolute values.

Perform pruning by 10%, 20%, 30%, 40%, and 50% of the weights.

Measure the performance of the network before and after retraining. Report the F1 score on both the training and testing sets and compare the results with the precise (non-pruned) network.

4.2 Data compression

Task 2: Apply PCA on the MNIST dataset and compress the images by removing 20% and 40% of the least important components. For this purpose, flatten the images, apply PCA, remove the selected components, and reconstruct the images back to their original size.

Retrain the CNN model with the compressed dataset and measure its performance on the test set.

Task 3: (Remove non-important input pixels): Remove the non important pixels that have the lowest variance in the given input data by 20% and 40%. The variance should be computed for each pixel location across the entire training dataset.

Retrain the CNN model with the compressed dataset and measure its performance on the test set.

Task 4: Select one task among Task 2 and Task 3 and combine it with Task 1. First apply the selected data compression technique and then perform weight pruning on the compressed dataset.

Report the performance of the network and compare the results with the previous tasks.

4.3 Report

For each given task, you should provide a brief explanation about the results you obtained. For better understanding, a tabular representation of results is preferable. If you use the jupyter notebook, you may write your report in the jupyter notebook.

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